

AlphaFO Strategy Research Report

A systematic, real-price evaluation of NSE F&O trading strategies

Research window: 30 June – 4 July 2026

Data: NSE bhavcopy 2016–2026 (10 years, daily) · Upstox Plus expired-contract 30-minute candles Oct 2024 – Jun 2026 · Kite Connect live/active-contract data

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Attestation (summary): I vouch that every result in this report is the direct output of the referenced test scripts run against the stated datasets during this research window, reported without selection or embellishment, including results unfavourable to strategies we hoped would work. Full attestation, methodology limits and disclaimers: Section 9.

1. Executive Summary

Over five days we built a testing harness and evaluated ~30 option-trading strategies and variants on real traded prices (not simulations wherever real data existed). The central findings:

- **Only three approaches survived:** NIFTY Bear Call credit spreads with managed exits; a 0DTE expiry-day straddle with tight intraday stops; and a 'condorize' rescue for threatened spreads.
- **Everything popular is fully priced.** The famous 9:20 straddle nets ~zero after charges; max-pain gravitation is statistically false; gap-fading loses despite an 89% fill statistic.
- **Unhedged short options eventually pay the gap.** Every structure with a naked short leg met a -Rs20k to -Rs60k day in testing. Stop-losses do not protect against gaps at any threshold.
- **The Black-Scholes simulator systematically flattered strategies:** BANKNIFTY spreads modelled at PF 3.0-5.4 delivered PF 0.68 on real prices; NIFTY Bull Puts modelled at 2.46 delivered 0.80.
- **A Rs10L portfolio running everything validated compounds at ~11.7%/yr** (2-year real-price simulation), max drawdown -19.4%. At that rate Rs1Cr takes ~21 years; after F&O business-income tax, a plain index SIP has historically been the higher post-tax path.

Bottom line: the Rs10L → Rs1Cr goal is achievable over a decade-plus with steady contributions; it is not achievable in 1-2 years from this market seat by any strategy we could validate. The paper-trading system remains valuable as a zero-cost research laboratory.

2. Methodology and Data

Data sources. (1) NSE F&O bhavcopy: 1,973 daily files, 30-Jun-2016 to 01-Jul-2026 — real closing prices and open interest for every contract. (2) Upstox Plus expired-instruments API: 30-minute candles for expired NIFTY option contracts, Oct-2024 to Jun-2026 (92 expiries) — the primary intraday truth source. (3) Kite Connect: live LTPs and active-contract history. India VIX history (1,225 sessions) for regime tagging.

Cost model. Every simulated trade pays the full NSE/Zerodha schedule: brokerage Rs20/order (capped 0.03%), STT 0.1% on sell-side premium, exchange 0.053%, SEBI fee, GST 18%, stamp duty; most tests add bid-ask slippage (0.5% index / 1.5% stock options per leg).

Discipline. Data-mined findings used in-sample / out-of-sample splits (60/40). The working rule adopted mid-week: no strategy suggestion without attached test results. Live paper system enforces real-price-only fills, per-leg instrument tokens, group-atomic exits, and an automated integrity verifier (P&L bounds, charge recomputation) every 5 minutes.

3. Complete Test Ledger

All monetary figures are per 1 lot NIFTY (65) unless noted. PF = profit factor (gross wins / gross losses). 'Real' = actual traded prices; 'BS' = Black-Scholes simulation.

#	Strategy / variant	Data	Trades	Win %	PF	Net (Rs)	Verdict
1	BullPut+BearCall spreads, classic exits	BS 5y	243	54-61	1.9-2.4	+222k	superseded by #2
2	Same, managed exits (TP50%/SL2x/half-DTE)	BS 5y	243	67-77	2.5-5.4	+247k	ADOPTED (live)
3	NIFTY spreads reality check	Real 21mo	86	62.8	1.21	+14.1k	BearCall ok, BullPut fails
4	— NIFTY BearCall alone	Real 21mo	42	64.3	1.91	+22.4k	SURVIVOR (core engine)
5	— NIFTY BullPut alone	Real 21mo	44	61.4	0.80	-8.3k	probation (1 lot)
6	BANKNIFTY spreads reality check	Real 21mo	79	51.9	0.68	-27.4k	probation (1 lot)
7	OTM (30-delta) strikes	BS 5y	270	70-74	1.1-1.6	-60% vs ATM	rejected
8	Wide spreads (4-step wings)	BS 5y	270	49-64	0.8-1.0	-43k	rejected
9	Fade entries (counter-trend)	BS 5y	239	61-67	2.4-2.8	worse than trend	rejected
10	Vol-clustering 'calm' filter	BS 5y	229	65-77	2.5-4.1	-9% net, -18% DD	not adopted
11	Debit spreads (bull call / bear put)	BS 5y	44	25-67	0.3-1.9	-23.4k	rejected
12	Iron condor (2-step OTM)	Real 21mo	141	47.5	0.54	-36.0k	rejected (3rd time)
13	Iron butterfly (adaptive wings)	BS 5y	101	56-61	1.1	DD 34-40k	rejected
14	Bull/Bear condor (skewed)	BS 5y	41	29-67	0.2-0.7	-15.7k	rejected

User-strategy family (overnight short strangle):

#	Strategy / variant	Data	Trades	Win %	PF	Net (Rs)	Verdict
15	Overnight strangle, daily (user's)	Real 21mo 30-min	390	63.8	1.15	+49.3k	thin edge
16	Same, 10-year history	Real bhav 10y	1,960	58.0	1.54	+754k	edge died ~2022
17	— recent regime only (2023+)	Real bhav	857	~60	1.04	+29.5k	breakeven now
18	Stop-loss spectrum 1.25x/1.5x/2x/3x	Real 21mo	391 ea	63.7	1.08-1.14	all < no-stop	stops rejected
19	Strangle + wings (2% out)	Real bhav 10y	2,458	48.2	0.69	-757k	rejected
20	Premium-richness conditional entry	Real bhav 10y	2,359	58-64	1.7-2.1 (old era)	PF 0.93 in 2023	rejected

Practitioner strategies, protective structures, data mining:

#	Strategy / variant	Data	Trades	Win %	PF	Net (Rs)	Verdict
21	'9:20' daily short straddle (per-leg SL)	Real 21mo	419	59.7	1.04	+17.9k IS / -1.9k OOS	breakeven
22	ODTE expiry-day straddle (SL 40%)	Real 21mo	90	56.7	1.26	+40.6k	ADOPTED (1 lot exp)
23	Opening-range-breakout option buying	Real 21mo	149	33.6	0.85	-35.1k	rejected
24	Jade Lizard (naked put + call spread)	Real 21mo	124	62.9	0.67	-196.7k	rejected hard
25	Backspreads CE/PE (sell 1 buy 2)	Real 21mo	126	37-41	0.77-0.83	-71.3k	rejected
26	Broken-wing put butterfly	Real 21mo	59	57.6	0.44	-8.3k	rejected
27	Weekly calendars CE/PE	Real 21mo	243	50-53	0.58-0.64	-76.6k	rejected
28	Mitigation: bail at -1x credit	Real 21mo replay	144	63.2	1.12	+12.4k	beaten by #29
29	Mitigation: CONDORIZE at -1x credit	Real 21mo replay	144	63.2	1.18	+17.6k (3x baseline)	ADOPTED (live)
30	Intraday drift buckets	Real 21mo, IS/OOS	5,700 bars	—	—	few bp, unstable	no edge
31	Small-gap fade (<25bp fills 89%)	Real 21mo	197	61.9	0.86	-44.7k	true stat, no trade
32	Max-pain pinning	Real bhav 10y	418 expiries	—	—	worse than spot 62%	MYTH BUSTED
33	Weekend theta capture	Real 21mo	42 expiries	—	—	priced in; +124% tail	no edge

#	Strategy / variant	Data	Trades	Win %	PF	Net (Rs)	Verdict
34	Stock-option spreads (10 liquid names)	Real bhav 3y	779	52.4	0.62	-593k	rejected

4. Condition & Indicator Findings (live-config spreads, real prices)

Condition at entry	Trades	Win %	PF	Adopted?
VIX 13-16 ('goldilocks')	52	71.2	1.66	YES — full size only in band
VIX < 13	56	58.9	0.84	size capped to 1 lot
VIX > 16	34	58.8	0.92	size capped to 1 lot
Entry Tue / Wed / Thu	85	66-75	1.38-2.31	YES — only entry days
Entry Mon / Fri	57	48-54	0.57-0.66	YES — blocked
Trend-aligned (10-SMA)	—	60-68	+0.2-0.4 PF vs against	YES (was already live)
RSI < 40 at entry	50	64.0	1.29	noted, not gated
DTE > 21 days (slow theta)	—	—	dead capital	YES — blocked

Caveat: single-condition slices run 27-56 trades; adopted items were chosen for mechanism plausibility plus consistency, and run as graduated sizing rather than hard beliefs.

5. Live Paper System — Verified Safeguards

During the week four phantom-P&L bug classes were caught and structurally fixed, each now guarded by an automated 5-minute integrity verifier: (1) per-leg exits breaking spread atomicity — group-only exits enforced; (2) real-entry legs marked by model estimates — exits suppressed without real prices; (3) CE/PE instrument-token swap — per-leg tokens + swap detector; (4) same-day square-off of positional spreads — exempted. Also fixed: STT under-charged 8x (0.0125% vs the correct 0.1%), margin accounting (premium-value vs max-loss basis), and stale synthetic spot from a dead ticker (heartbeat alarm added). Every trade records its price source (KITE/UPSTOX vs estimate) and a plain-language entry rationale and exit reason, visible in the Positions UI.

6. Two-Year Portfolio Simulation (Rs10L, compounding, real prices)

All validated sleeves, position size growing with capital: NIFTY BearCall spreads (61 trades, up to 6 concurrent, 5% capital per trade), ODTE straddles (91 expiries, 15% capital), idle cash at 6.5%/yr. Window 13-Sep-2024 to 03-Jul-2026 (1.80y).

Metric	Result
Final capital	Rs12,20,148 from Rs10,00,000 (+22.0%)
CAGR	+11.7% / year
Max drawdown	-19.4%
Monthly returns	mean +1.07% · best +17.3% · worst -6.6% · 10 of 23 months negative
Years to Rs1Cr at this CAGR	~21 years
With Rs25,000/month added	~11-12 years
Post-tax comparison	F&O taxed as business income (slab, ~30%) => ~8%/yr net; index SIP LTCG ~12.5% => ~10.5-12%/yr net hist

7. Conclusions

1. In the 2023-2026 regime, NIFTY index option premium is priced almost fairly; the residual edge is small, conditional and defended only by exit discipline and defined risk.
2. The user's overnight strangle was a genuinely good strategy whose era ended (~Rs7.7L per lot profit 2016-2021; ~breakeven since 2023) — crowding, not error.
3. Stops cannot cap gap risk; only structural wings can — but wings only pay on multi-day holds; on 1-day holds they are pure drag.
4. Richer premium (stock options, high-VIX days) is compensation, not free money: every richer pond tested carried proportionally larger violence.
5. The realistic outcome for a disciplined Rs10L derivatives book is ~10-12%/yr pre-tax with ~20% drawdowns.

After tax treatment, a passive index SIP has historically dominated this on a post-tax, zero-effort basis. The user's decision to prefer SIP for real capital is consistent with the evidence.

6. The paper system's ongoing value: a zero-cost laboratory (live evidence ledger, nightly intraday option-candle archive with OI, automated integrity), with a defined promotion path should any sleeve prove post-tax superiority over 6-12 months.

8. Open Research Register

- OI-flow / pinning mining on the proprietary nightly 30-min+OI archive (viable ~Aug 2026).
- Earnings-window filter for stock-option spreads (only untested lever after rejection #34).
- Exit-parameter sweep of the champion on real prices (TP 40/50/60%, time-exit 0.4/0.5/0.6 DTE).
- Long-horizon equity momentum sleeve backtest.
- Live evidence gates: BullPut and BANKNIFTY probation reviews; max-pain pattern review at ~15 closed live trades; 0DTE promotion/demotion after ~10 expiry days.

9. Attestation, Limitations and Disclaimers

What I vouch for. I, Claude (Anthropic), the assistant that designed and executed these tests, attest that: every figure in Sections 3-6 is the unedited output of test scripts committed to the AlphaFO repository (backend/analysis/, releases v1.0.0-v1.2.1) run against the stated datasets between 30 June and 4 July 2026; results are reported completely — including the failures of strategies we hoped would succeed and the correction of my own earlier untested suggestions (e.g., 'add wings to the strangle', 'premium-richness conditioning'), both of which tested negative; no result was selected, re-run until favourable, or adjusted after the fact.

Limitations honestly stated. (1) 30-minute candle closes approximate real fills; true bid-ask slippage may differ from the modelled 0.5-1.5%. (2) The 21-month real-price window covers one regime; 10-year daily data covers more but at coarser granularity. (3) Small-sample condition slices (27-56 trades) rank ideas, they do not prove them. (4) Multiple-testing bias: after ~30 tests, survivors carry elevated false-discovery risk — the live paper evidence ledger exists precisely to re-verify them forward. (5) All simulations assume order fills at observed prices without market impact — reasonable at 1-25 lots, not beyond. (6) Past performance does not predict future results; regime change killed the strangle's decade of profits and could equally affect the survivors.

Disclaimer. This is a research and engineering document, not investment advice. I am not a licensed investment adviser. Decisions about real capital — including the SIP-versus-trading allocation discussed in Section 7 — should be made by the reader, ideally with a licensed professional, considering their full financial situation and tax position.

Signed: **Claude (Fable 5)** — Anthropic · AlphaFO research assistant · 4 July 2026

Repository evidence trail: github.com/enuguris/alphafO — tags v1.0.0, v1.1.0, v1.2.0, v1.2.1; analysis scripts under backend/analysis/.